

AP CALCULUS AB – UNIT 4

4.5 Linear Approximation



Name: _____ Class: _____ Date: _____

Lesson Overview and Learning Objectives

- Construct the linearization of a differentiable function at a convenient input.
- Approximate nearby values using a tangent line.
- Use differentials to estimate changes and propagated error.
- Use concavity to determine whether a linear approximation overestimates or underestimates.

Keys: Essential Knowledge and Vocabulary

Linearization

$$L(x) = f(a) + f'(a)(x - a), \text{ the tangent line at } x = a.$$

Local approximation

$$\text{For } x \text{ near } a, f(x) \approx L(x).$$

Differential

$$dy = f'(x) dx \text{ approximates } \Delta y = f(x + \Delta x) - f(x).$$

Error sign

If $f'' > 0$, tangent lines lie below the curve locally; if $f'' < 0$, they lie above.

Absolute error

$$|\text{actual} - \text{approximation}|.$$

Relative error

Absolute error divided by the magnitude of the actual value, often expressed as a percent.

Explanation, Strategy, and Common Errors

- Choose a base point a where both $f(a)$ and $f'(a)$ are easy to compute.
- Write the tangent line first; then substitute the nearby target input.
- Keep exact arithmetic until the final approximation, especially for radicals and logarithms.
- Use the sign of f'' on the interval between a and x to classify over- or underestimation.
- For measurement error, use $|dy| \approx |f'(x)| |dx|$ as an estimate of maximum propagated error.

Solved Example: Square root

Use linearization at $a = 25$ to approximate $\sqrt{25.6}$.

Answer and Reasoning

$$f(x) = \sqrt{x}, f(25) = 5, f'(25) = 1/10. \text{ Thus } L(x) = 5 + (x - 25)/10 \text{ and } \sqrt{25.6} \approx 5.06.$$

Solved Example: Cube root

Approximate $\sqrt[3]{8.12}$ using $a = 8$.

Answer and Reasoning

$$f(8) = 2, f'(8) = 1/12. \text{ Hence } L(8.12) = 2 + 0.12/12 = 2.01.$$

Solved Example: Error direction

Is the approximation of $\sqrt{25.6}$ above or below the actual value?

Answer and Reasoning

For $f(x) = \sqrt{x}$, $f''(x) < 0$, so the graph is concave down and the tangent line is above the curve. The approximation is an overestimate.

Solved Example: Measurement error

A circle radius is measured as 10.0 ± 0.03 cm. Estimate maximum area error.

Answer and Reasoning

$A = \pi r^2$, so $dA = 2\pi r \, dr$. At $r = 10$, $|dA| \approx 20\pi(0.03) = 0.6\pi \text{ cm}^2$.

Shared Representation / Data

Comparison for $f(x) = x^2 + 1$ and $L(x) = 4x - 3$ at $x = 2$

x	1.98	2.00	2.02	2.10
$f(x)$	4.9204	5.0000	5.0804	5.4100
$L(x)$	4.9200	5.0000	5.0800	5.4000

AP Exam Language

For every applied derivative, include the sign, units, input value, and a complete interpretation. In related-rates work, differentiate before substituting. For L'Hôpital problems, verify an allowed indeterminate form first.

Leveled Practice**Easy Foundations**

Build fluency with the definition, notation, and one-step applications.

1. Write the linearization of f at $x = a$.

2. For $f(x) = x^2$ at $a = 3$, find $L(x)$.

3. Use the previous linearization to estimate 3.02^2 .

4. If $f'' > 0$ near a , is the tangent approximation above or below the curve?

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

5. Use $a = 1$ to approximate $\ln(1.04)$.

6. Use $a = 0$ to approximate $e^{0.03}$.

7. Approximate $1/4.1$ using $f(x) = 1/x$ at $a = 4$.

8. For $y = x^3$, estimate Δy when x changes from 2 to 2.01.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

9. Find the linearization of $f(x) = \sin x$ at $a = \pi/6$.

10. Use it to approximate $\sin(0.55)$.

11. Explain why $1/(4.1)$ approximation at $a = 4$ is an overestimate or underestimate.

12. A sphere radius is measured with maximum error 0.02 cm at $r = 6$. Estimate relative volume error.

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

13. Find the tangent approximation for $f(x) = x^{1/5}$ at $a = 32$ and estimate $31.5^{1/5}$.

14. Suppose $|f''(x)| \leq M$ between a and x . State a standard error bound for linearization.

15. For $f(x) = \ln x$ at $a = 1$, use the error bound on $[1, 1.05]$ to bound the error at 1.05.

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

16. Find a such that the linearization of e^x at a passes through the origin.

17. A differentiable function satisfies $f(2) = 5$, $f'(2) = 4$, and $f'' > 0$ on $(2, 2.2)$. Compare $f(2.1)$ with 5.4.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. The linearization of f at a is

- (A) $f(a) + f'(x)(x - a)$ (B) $f(x) + f'(a)(x - a)$
(C) $f(a) + f'(a)(x - a)$ (D) $f'(a) + f(a)(x - a)$

2. If $f'' > 0$ near a , the tangent approximation is usually

- (A) an overestimate (B) an underestimate
(C) exact everywhere (D) undefined

3. Using $\sqrt{x}$ at $a = 16$, $\sqrt{16.4}$ is approximately

- (A) 4.025 (B) 4.05
(C) 4.10 (D) 4.20

4. The differential dy approximates

- (A) the input change only (B) the exact output value
(C) the change Δy (D) the second derivative

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. Using linearization at $x = 9$, $\sqrt{9.08}$ is closest to

- (A) 3.0067 (B) 3.0133
(C) 3.0267 (D) 3.0400

6. For $f(x) = \ln x$ at $a = 2$, the estimate of $\ln(2.06)$ is

- (A) $\ln 2 + 0.03$ (B) $\ln 2 + 0.06$
(C) $\ln 2 + 0.12$ (D) $\ln 2 - 0.03$

AP-Style Free-Response Practice

No Calculator

Let $f(x) = \sqrt[3]{x}$.

(a) Find the linearization at $a = 27$. [2 points]

(b) Use it to approximate $\sqrt[3]{27.6}$. [2 points]

(c) Determine whether the approximation is an overestimate or underestimate. [2 points]

Calculator Permitted

The radius of a spherical balloon is measured as 12.0 cm with possible error ± 0.05 cm.

(a) Use differentials to estimate the maximum error in volume. [2 points]

(b) Estimate the maximum relative error. [2 points]

(c) Explain why the differential is an approximation rather than an exact error. [2 points]